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1. COMPANY PROFILE

ARDENT (Ardent Computech Pvt. Ltd.), formerly known as Ardent Computech Private Limited, is an ISO 9001:2015 certified Software Development and Training Company based in India. Operating independently since 2003, the organization has recently undergone a strategic merger with ARDENT Technologies, enhancing its global outreach and service offerings.

ARDENT Technologies

ARDENT Technologies delivers high-end IT services across the UK, USA, Canada, and India. Its core competencies lie in the development of customized application software, encompassing end-to-end solutions including system analysis, design, development, implementation, and training.The company also provides expert consultancy and electronic security solutions. Its clientele spans educational institutions, entertainment companies, resorts, theme parks, the service industry, telecom operators, media, and diverse business sectors.

ARDENT Collaborations

ARDENT Collaborations, the Research, Training, and Development division of ARDENT (Ardent Computech Pvt. Ltd.), offers professional IT-enabled services and industrial training programs .These programs are tailored for freshers and professionals from B.Tech, M.Tech, MBA, MCA, BCA, and MSc backgrounds. ARDENT (Ardent Computech Pvt. Ltd.) provides Summer Training, Winter Training, and Industrial Training to eligible candidates.High-performing students may qualify for stipends, scholarships, and additional benefits based on performance and mentor recommendations.

Associations and Accreditations

ARDENT (Ardent Computech Pvt. Ltd.) is affiliated with the National Council of Vocational Training (NCVT) under the Directorate General of Employment & Training (DGET), Ministry of Labour & Employment, Government of India. The institution upholds strict quality standards under ISO 9001:2015 certification and is dedicated to bridging the gap between academic knowledge and industry skills through innovative training programs.

2. INTRODUCTION

In the ever-evolving job market, traditional recruitment methods are often inefficient and time-consuming for both job seekers and recruiters. The **AI-Based Smart Job Matching and Recruitment System** aims to streamline the hiring process through intelligent filtering, automation, and application tracking, providing an efficient digital platform for both candidates and employers.

This system is designed using the MERN stack (Mongo DB, Express.js, React.js, Node.js) and allows users to create profiles, browse job listings, apply for jobs, and manage applications seamlessly. Additionally, the system integrates AI-based features to enhance job recommendations, analyze user profiles, and improve candidate-job matching accuracy. On the other hand, recruiters can post job openings, view applicants, and filter candidates based on criteria such as skill sets, experience, and location.

By introducing automation and intelligent matchmaking powered by AI, this system minimizes manual effort, improves transparency, and ensures better hiring outcomes.

2A. Project Overview

The AI-Based Smart Job Matching and Recruitment System is a web-based application designed to simplify and modernize the recruitment process for both job seekers and recruiters. The system provides a centralized platform where users can create profiles, search for jobs, apply online, and track application status efficiently.

This project has been developed to overcome the limitations of traditional hiring methods, which are often time-consuming, manual, and inefficient. By leveraging modern web technologies such as the MERN stack (Mongo DB, Express.js, React.js, Node.js), the system ensures a fast, scalable, and user-friendly experience.

The primary purpose of developing this system is to automate the recruitment process, reduce human effort, and provide accurate job-candidate matching. It helps organizations identify suitable candidates quickly while enabling job seekers to find relevant opportunities based on their skills and preferences.

2B. Problem Statement

In the current job market, traditional recruitment systems face several challenges, including inefficiency, lack of automation, and poor matching between job requirements and candidate profiles.

Job seekers often apply to multiple positions without proper guidance, resulting in irrelevant applications. On the other hand, recruiters spend a significant amount of time manually reviewing resumes and shortlisting candidates, which delays the hiring process.

Additionally, existing systems lack intelligent filtering, real-time tracking, and personalized recommendations. Therefore, there is a need for an **AI-Based Smart Job Matching and Recruitment System** that can improve efficiency, accuracy, and transparency in the hiring process.

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2C. Objectives

The main objectives of the **AI-Based Smart Job Matching and Recruitment System** are as follows:

- To develop a user-friendly and responsive web interface
- To implement secure authentication and authorization mechanisms
- To automate the recruitment and job application process
- To provide accurate job recommendations based on user skills using AI
- To enable real-time application tracking for users
- To reduce manual effort in candidate shortlisting through intelligent filtering
- To improve communication between job seekers and recruiters
- To ensure efficient data management and system scalability

2D. Scope of the Project

The scope of this project includes the design and development of an **AI-Based Smart Job Matching and Recruitment System**, which is a web-based job portal that can be used by job seekers, recruiters, and administrators.

Job seekers can register, create profiles, search for jobs, and apply online. Recruiters can post job vacancies, manage applications, and filter candidates based on skills, experience, and location using intelligent and automated features.

The system can be used in various domains such as IT companies, corporate organizations, educational institutions, and recruitment agencies. It is accessible through web browsers and can be deployed on cloud platforms for wider accessibility and scalability.

Future enhancements may include advanced AI-based job recommendations, resume parsing, interview scheduling, and real-time communication features.

2E. Methodology

The development of this project follows a structured Software Development Life Cycle (SDLC) approach combined with iterative improvements.

The methodology includes the following steps:

• **Requirement Analysis:** Understanding user requirements such as job search, job posting, and application tracking in the **AI-Based Smart Job Matching and Recruitment System**.

• **System Design:** Designing system architecture, database schema, and user interface layout.

• **Development:** Frontend development using React.js and backend development using Node.js and Express.js.

• **Database Integration:** Using Mongo DB to store user data, job listings, applications, and AI-related data.

• **Testing:** Performing unit testing and integration testing using tools such as Postman and browser-based testing.

• **Deployment:** Deploying the application on local or cloud environments for real-time access.

• **Maintenance:** Updating the system, fixing bugs, and adding new features based on user feedback and AI enhancements.

This structured methodology ensures that the project is developed efficiently, with proper planning, testing, scalability, and integration of intelligent features.

3. LITERATURE REVIEW

The literature review provides an understanding of existing recruitment platforms, their working models, limitations, and the need for an improved system. In the field of online recruitment, many web-based job portals have been developed to connect job seekers with recruiters. These systems have significantly reduced the dependency on traditional hiring methods and introduced digital transformation in the recruitment process.

Several existing job portal systems such as LinkedIn, Naukri, Indeed, and Monster provide essential recruitment services, including profile creation, job searching, job posting, and online application submission. These platforms have improved accessibility and speed in the hiring process. However, many of them still depend on basic search and filtering mechanisms, which often lead to irrelevant job recommendations and inefficient candidate selection.

The proposed **AI-Based Smart Job Matching and Recruitment System** is designed to address these gaps by introducing a more efficient, user-friendly, and intelligent recruitment platform.

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3A. Existing Systems

Existing online recruitment systems are web-based platforms that allow job seekers to search for job opportunities and apply online, while recruiters can post vacancies and review candidate profiles. These systems have replaced many traditional recruitment practices such as newspaper advertisements, consultancy-based hiring, and manual resume collection.

Popular job portals provide the following common features:

- User registration and profile creation
- Job search by title, company, or location
- Online job application submission
- Recruiter dashboard for posting vacancies
- Resume upload and application tracking

Although these systems have improved the recruitment process, most of them provide generalized search results rather than personalized and accurate job recommendations. They usually focus on listing a large number of jobs instead of ensuring proper matching between candidate skills and job requirements.

These limitations highlight the need for an **AI-Based Smart Job Matching and Recruitment System**, which can provide intelligent filtering, personalized recommendations, and more accurate candidate-job matching.

3B. Limitations of Existing Systems

Despite their popularity and usefulness, existing recruitment systems have several limitations that affect their efficiency and user experience.

The major limitations are as follows:

• **Lack of Intelligent Job Matching:** Most systems do not provide accurate matching between job seeker profiles and job requirements.

• **Excessive Manual Screening:** Recruiters often need to manually review a large number of applications, which is time-consuming.

• **Irrelevant Job Recommendations:** Job seekers frequently receive job suggestions that do not fully match their skills, experience, or preferences.

• **Limited Personalization:** Existing systems often fail to customize job recommendations based on user behavior or profile data.

• **No Proper Real-Time Tracking:** Many portals do not provide clear and real-time updates regarding application status.

• **High Competition and Data Overload:** Users face difficulties in identifying suitable jobs from a large volume of listings.

These limitations show that there is still a need for a more advanced and efficient recruitment platform, such as an **AI-Based Smart Job Matching and Recruitment System**, which can provide intelligent filtering, personalized recommendations, and improved efficiency.

3C. Comparison with the Proposed System

The proposed **AI-Based Smart Job Matching and Recruitment System** offers several improvements over traditional and existing online job portals. It is designed to provide a more intelligent, transparent, and efficient recruitment experience for both job seekers and recruiters.

The comparison is as follows:

System Type	Features	Advantages	Limitations
Traditional Recruitment System	Manual hiring, paper resumes, offline interviews	Simple and familiar process	Slow, time-consuming, and less efficient
Existing Online Job Portals	Online job search, profile creation, job application	Easy access and wider reach	Limited personalization and weak matching accuracy
Proposed AI-Based Smart Job Matching and Recruitment System	Intelligent filtering, profile-based matching, application tracking, secure login	Faster hiring, better matching, improved user experience	Can be enhanced further with advanced AI and analytics

The proposed system provides a more structured and effective approach to recruitment. It reduces manual effort, improves job relevance, and helps recruiters shortlist suitable candidates more efficiently.

3D. Conclusion of Literature Review

From the study of existing recruitment systems, it is clear that web-based job portals have already improved the hiring process to a great extent. However, most systems still suffer from limitations such as poor matching accuracy, lack of personalization, and insufficient automation.

The proposed **AI-Based Smart Job Matching and Recruitment System** aims to overcome these issues by offering a more intelligent and user-friendly platform. Therefore, this project is relevant, practical, and necessary in the present competitive job market.

4. SYSTEM ANALYSIS

System Analysis is a crucial phase in software development that involves understanding system requirements, identifying user needs, and evaluating the feasibility of the proposed system. This phase ensures that the developed system meets user expectations and performs efficiently in real-world scenarios.

In the context of the **AI-Based Smart Job Matching and Recruitment System**, System Analysis helps in identifying the functional requirements, system behavior, and performance expectations, ensuring that the platform delivers accurate job matching, efficient recruitment processes, and a user-friendly experience.

4A. Requirement Analysis

Requirement Analysis defines the functionalities and performance expectations of the system. It is divided into Functional and Non-Functional Requirements.

Functional Requirements

Functional requirements describe the core features and operations of the **AI-Based Smart Job Matching and Recruitment System**.

The main functional requirements are as follows:

• **User Registration and Login:** The system must allow job seekers and recruiters to register and log in securely using their credentials.

• **User Profile Management:** Users should be able to create, update, and manage their profiles, including personal details, skills, and experience.

• **Job Posting:** Recruiters must be able to post, update, and delete job vacancies.
• **Job Search and Filtering:** Job seekers should be able to search for jobs based on keywords, location, and skills, with intelligent filtering support.

• **Job Application:** Users must be able to apply for jobs directly through the system.
• **Application Tracking:** Job seekers should be able to track the status of their applications in real time.

• **Candidate Management:** Recruiters should be able to view and filter applicants based on specific criteria.

• **Authentication and Security:** The system should ensure secure login using authentication methods such as JWT.

Non-Functional Requirements

Non-functional requirements define the quality and performance attributes of the system. The main non-functional requirements are:

- **Performance:**

  The system should respond quickly and handle multiple users efficiently.

- **Scalability:**

  The application must be able to handle increasing users and data without performance degradation.

- **Security:**

  User data must be protected using encryption, secure authentication, and authorization mechanisms.

- **Usability:**

  The interface should be user-friendly and easy to navigate.

- **Reliability:**

  The system should function correctly without frequent failures.

- **Availability:**

  The system should be accessible at all times with minimal downtime.

- **Maintainability:**

  The code should be structured and modular for easy updates and maintenance.

4B. Feasibility Study

The proposed system is technically feasible as it uses modern and widely adopted technologies such as the MERN stack (Mongo DB, Express.js, React.js, Node.js) for developing the **AI-Based Smart Job Matching and Recruitment System**.

These technologies provide:

• High performance and scalability
• Easy integration between frontend and backend
• Strong community support
• Cloud deployment capability

The system can be developed using standard tools such as Visual Studio Code, Postman, and Mongo DB Atlas, making implementation straightforward and efficient.

Economic Feasibility

The system is economically feasible as it requires minimal cost for development and deployment.

- Most technologies used (React, Node.js, Mongo DB) are open-source
- Development can be done using free tools
- Cloud services offer free or low-cost hosting options
- Maintenance cost is low due to modular structure

Thus, the overall cost of developing and maintaining the system is affordable.

Operational Feasibility

The system is operationally feasible as it is designed to be user-friendly and easy to use for both job seekers and recruiters in the **AI-Based Smart Job Matching and Recruitment System**.

- Simple registration and login process
- Easy navigation and job search features with intelligent assistance
- Minimal training required for users
- Efficient workflow for job application and recruitment

The system improves productivity by reducing manual work and enhancing the recruitment process through automation and AI-based features.

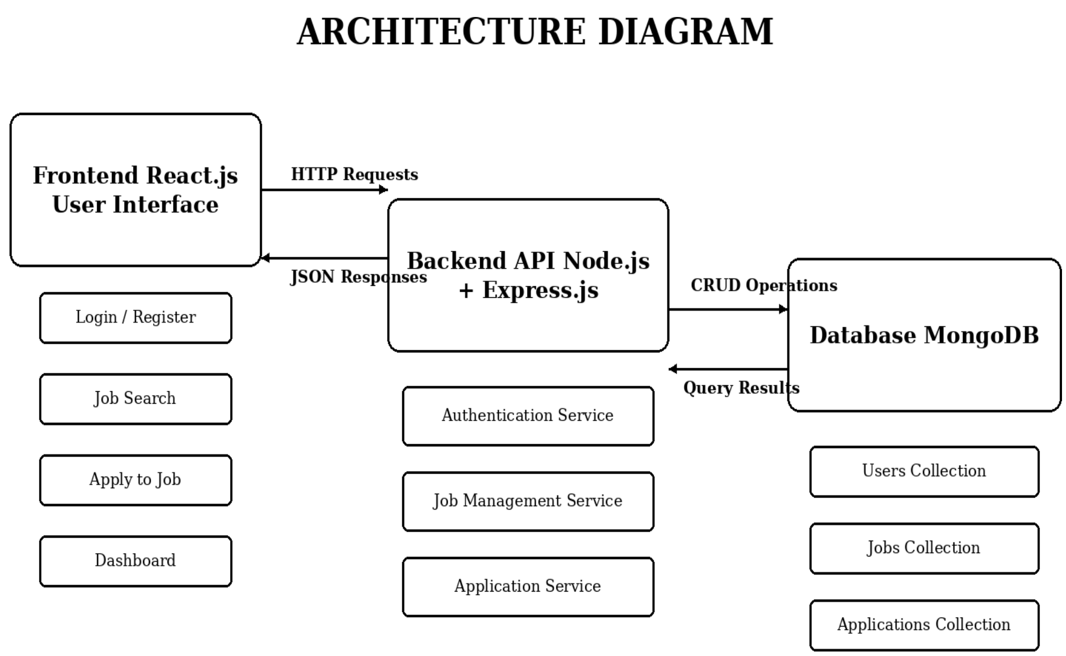
5. SYSTEM DESIGN

System Design defines the structural and behavioral representation of the **AI-Based Smart Job Matching and Recruitment System**. It presents the architecture of the application, the database design, and the UML diagrams required to explain system flow, user interaction, and process execution.

The following diagrams have been prepared in a clean academic format with properly aligned entities, clearly labeled relationships, and consistent arrow directions for accurate and professional major project documentation.

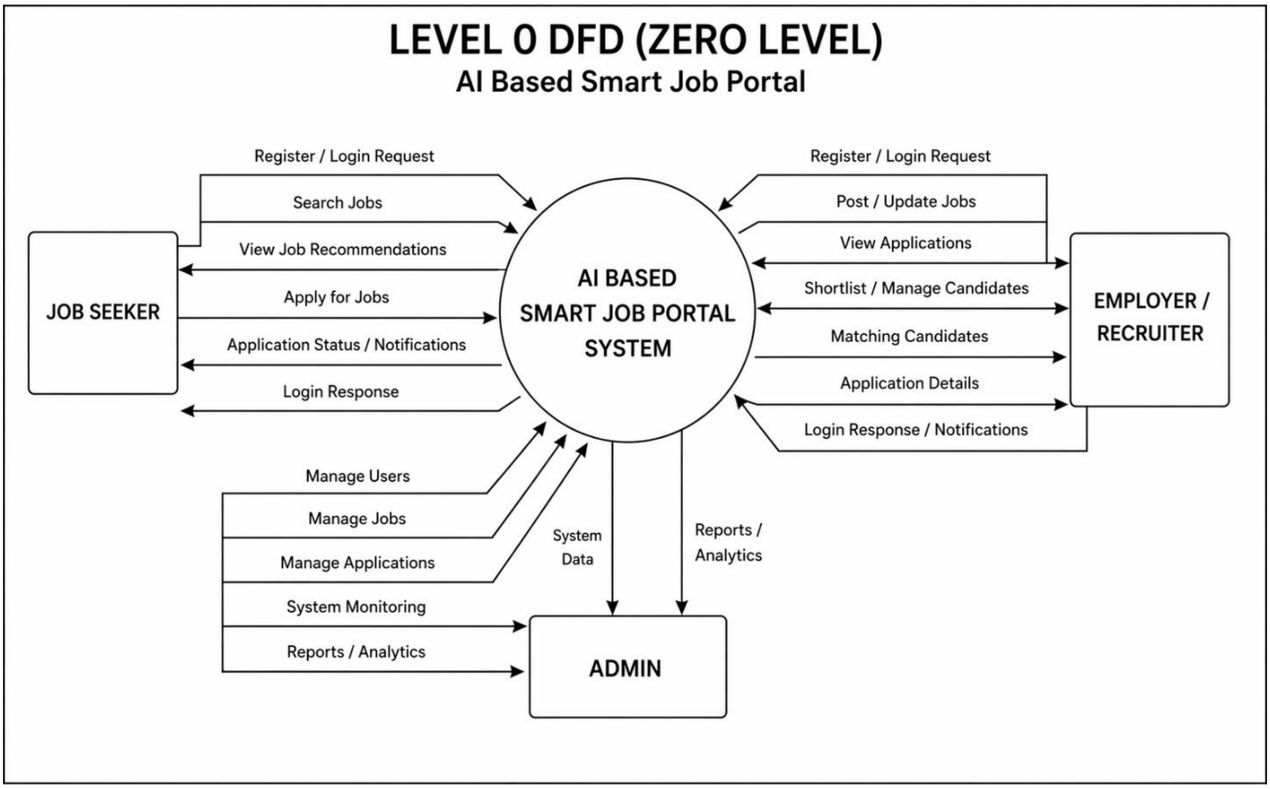
5A. **Architecture Diagram**

The architecture of the Job Portal Website follows a three-tier structure consisting of the Frontend, Backend, and Database layers. The frontend is developed using React.js to provide an interactive user interface. The backend uses Node.js and Express.js to process requests, manage authentication, handle job postings, and process job applications. Mongo DB is used as the database to store user, job, and application data.



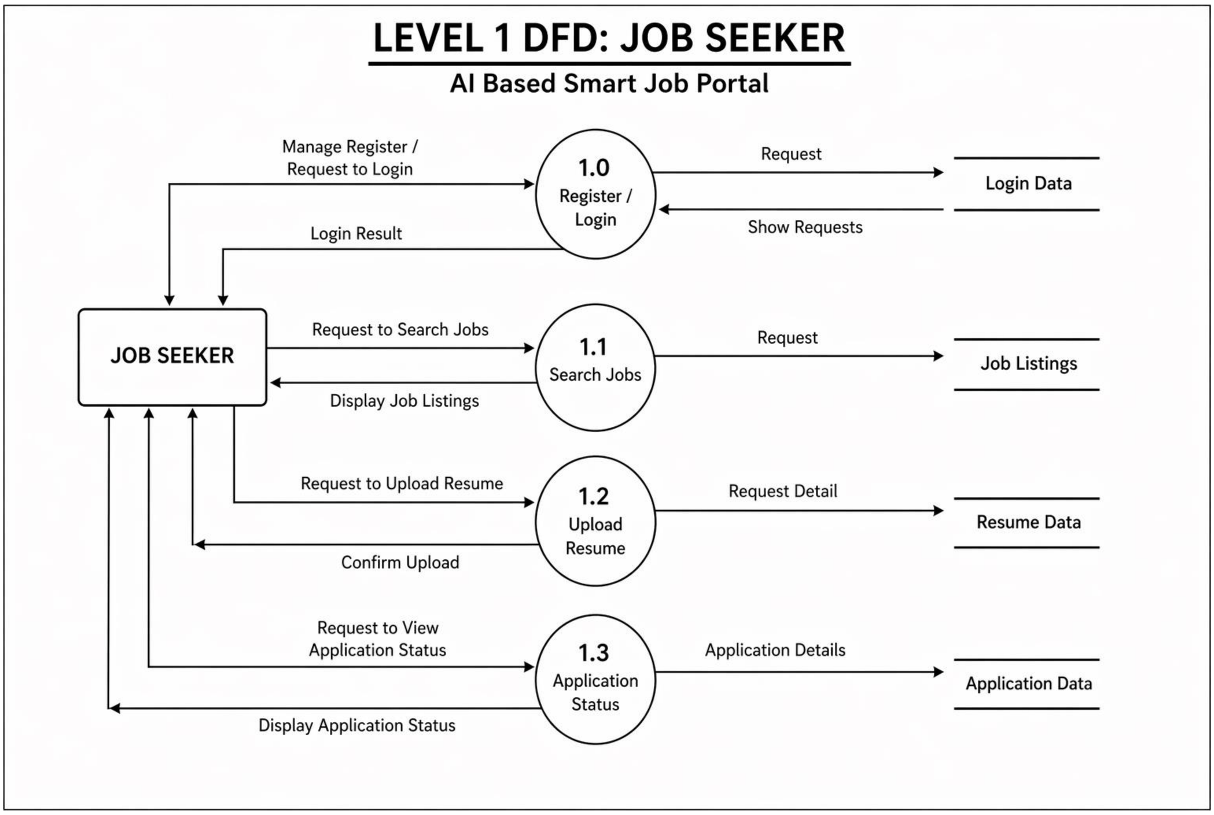
Data Flow Diagram (Level 0)

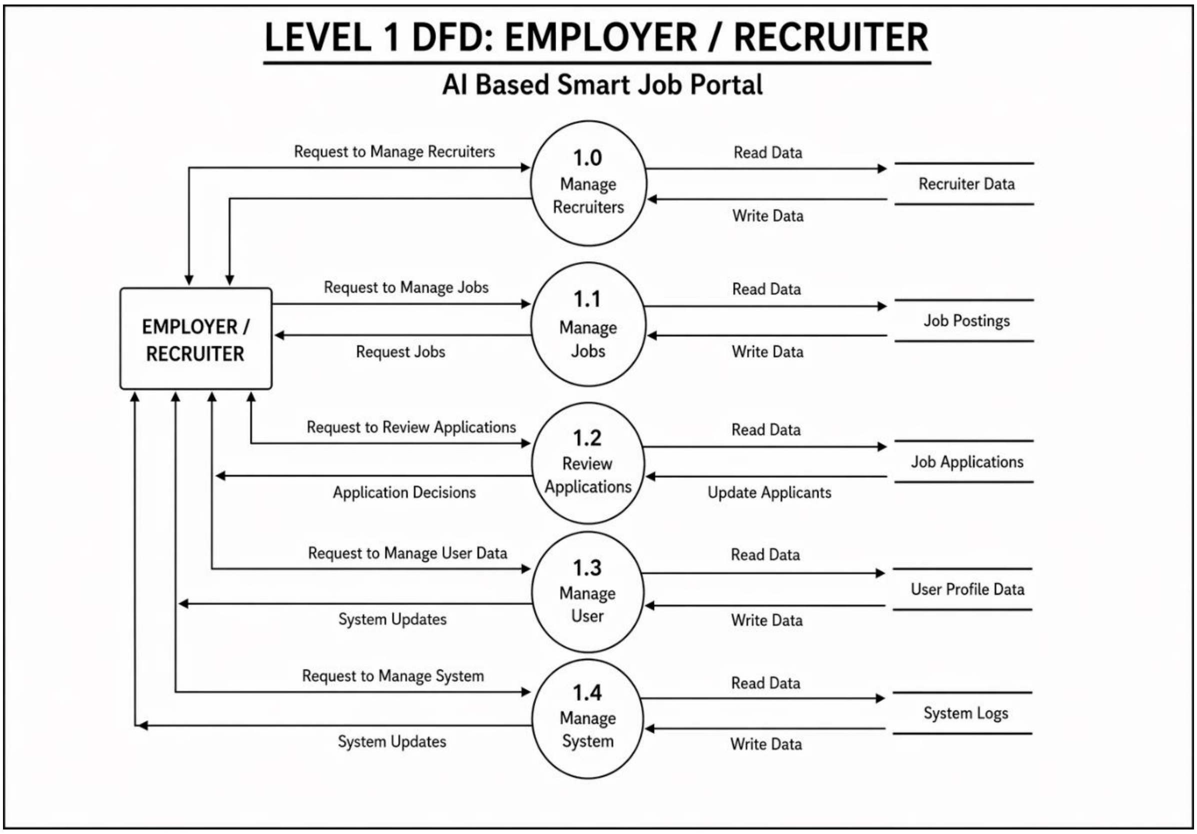
The Level 0 DFD shows the overall data flow between the external entities and the Job Portal System. It represents how the Job Seeker and Recruiter interact with the system for registration, login, viewing requests, and managing user-related information

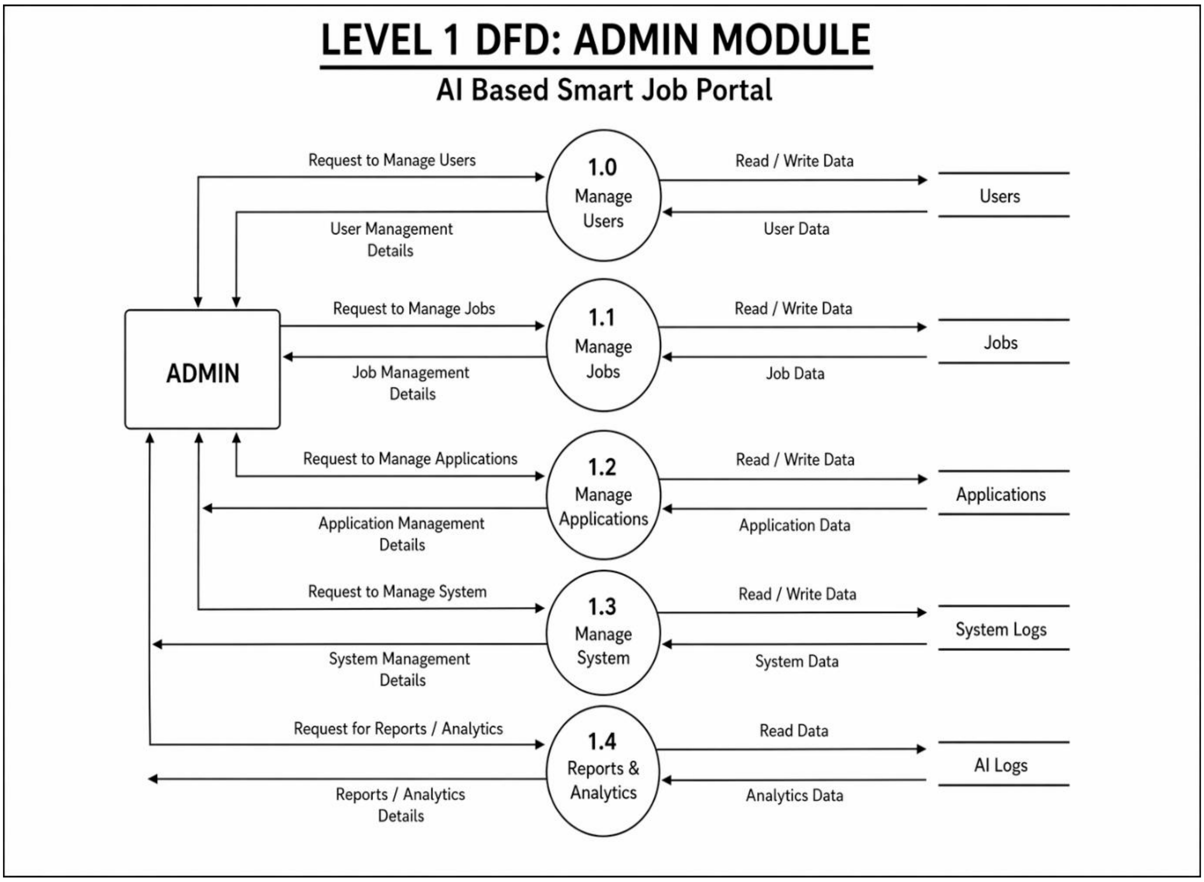


Data Flow Diagram (Level 1):

The Level 1 DFD expands the major functions of the **AI-Based Smart Job Matching and Recruitment System** for both Job Seeker and Recruiter roles. The Job Seeker flow includes registration/login, job search, resume upload, and application status tracking. The Recruiter flow includes recruiter management, job posting management, application review, user data management, and system management.

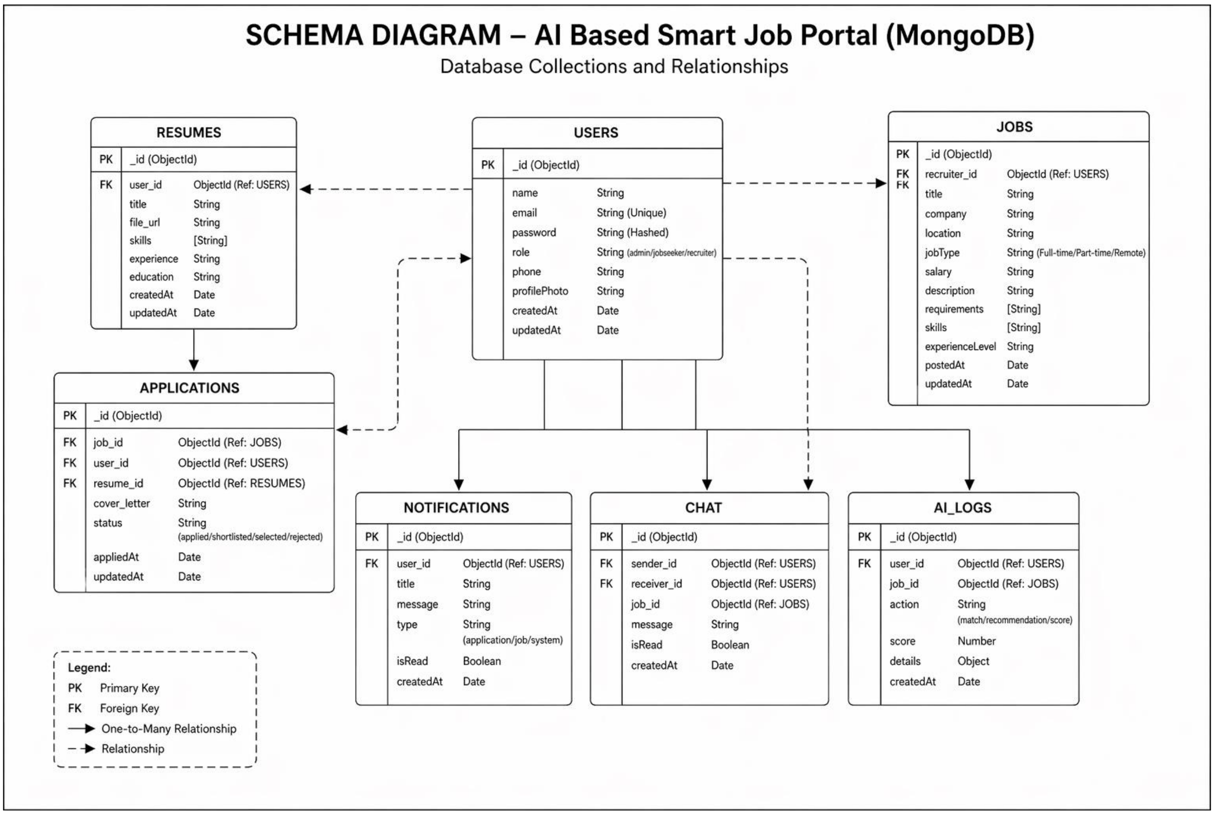



LEVEL 1 DFD: EMPLOYER / RECRUITER
AI Based Smart Job Portal
Request to Manage Recruiters
1.0
Manage
Recruiters
Read Data
Write Data
Recruiter Data
EMPLOYER / RECRUITER
Request to Manage Jobs
Request Jobs
1.1
Manage
Jobs
Read Data
Write Data
Job Postings
Request to Review Applications
Application Decisions
1.2
Review
Applications
Read Data
Update Applicants
Job Applications
Request to Manage User Data
System Updates
1.3
Manage
User
Read Data
Write Data
User Profile Data
Request to Manage System
System Updates
1.4
Manage
System
Read Data
Write Data
System Logs

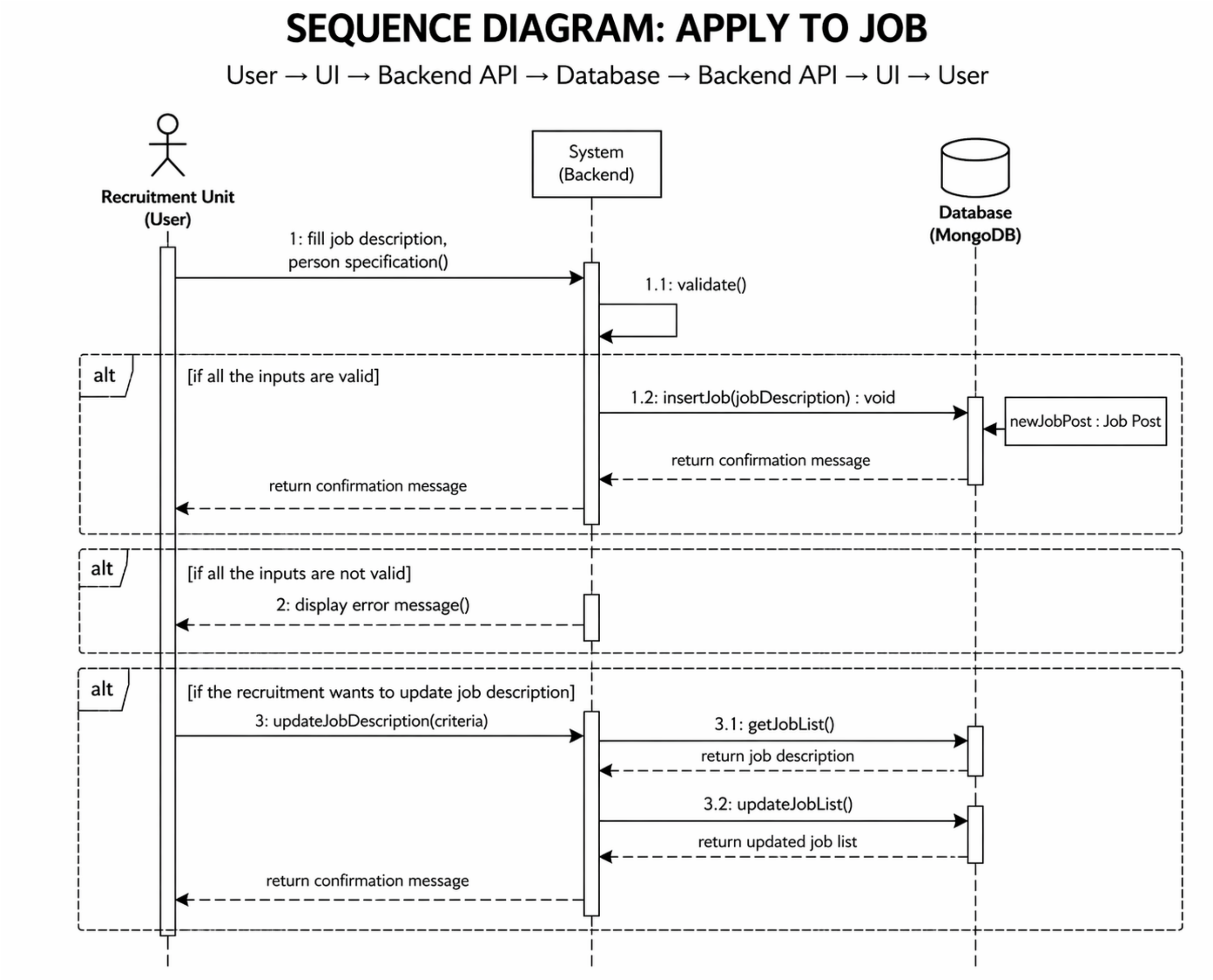

LEVEL 1 DFD: ADMIN MODULE
AI Based Smart Job Portal
Request to Manage Users
User Management Details
1.0
Manage
Users
Read / Write Data
User Data
Users
ADMIN
Request to Manage Jobs
Job Management Details
1.1
Manage
Jobs
Read / Write Data
Job Data
Jobs
Request to Manage Applications
Application Management Details
1.2
Manage
Applications
Read / Write Data
Application Data
Applications
Request to Manage System
System Management Details
1.3
Manage
System
Read / Write Data
System Data
System Logs
Request for Reports / Analytics
Reports / Analytics Details
1.4
Reports &
Analytics
Read Data
Analytics Data
AI Logs

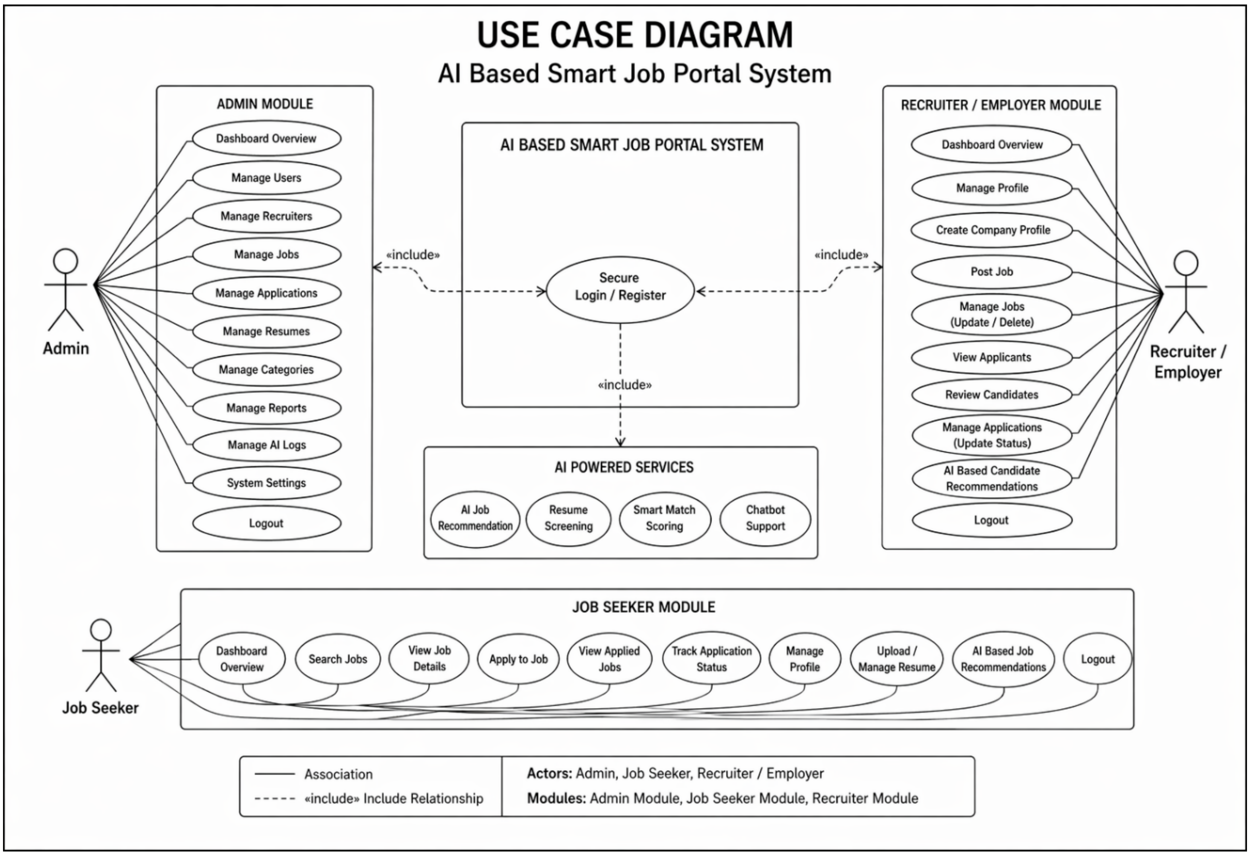
5B. Database Design

The database design of the Smart Job Matching and Recruitment System is based on Mongo DB collections. The design stores user profile data, job posting data, and application data in a structured way to maintain consistency, scalability, and efficient query processing.



5C. UML Diagrams




USE CASE DIAGRAM
AI Based Smart Job Portal System
ADMIN MODULE
Dashboard Overview
Manage Users
Manage Recruiters
Manage Jobs
Manage Applications
Manage Resumes
Manage Categories
Manage Reports
Manage AI Logs
System Settings
Logout
Admin
AI BASED SMART JOB PORTAL SYSTEM
«include»
Secure Login / Register
«include»
«include»
AI POWERED SERVICES
AI Job Recommendation
Resume Screening
Smart Match Scoring
Chatbot Support
RECRUITER / EMPLOYER MODULE
Dashboard Overview
Manage Profile
Create Company Profile
Post Job
Manage Jobs (Update / Delete)
View Applicants
Review Candidates
Manage Applications (Update Status)
AI Based Candidate Recommendations
Logout
Recruiter / Employer
JOB SEEKER MODULE
Dashboard Overview
Search Jobs
View Job Details
Apply to Job
View Applied Jobs
Track Application Status
Manage Profile
Upload / Manage Resume
AI Based Job Recommendations
Logout
Job Seeker
Association
«include» Include Relationship
Actors: Admin, Job Seeker, Recruiter / Employer
Modules: Admin Module, Job Seeker Module, Recruiter Module

6. TECHNOLOGY STACK

The AI-Based Smart Job Matching and Recruitment System is developed using the MERN stack with AI integration to ensure scalability, responsiveness, and intelligent functionality. The selected technologies provide a strong foundation for building a modern job portal that supports user management, job posting, application tracking, and advanced AI-based features such as smart job recommendations, resume screening, and intelligent candidate analysis.

Frontend Technologies

The frontend of the system is developed using **HTML, CSS, JavaScript, and React.js**.

- **HTML** is used to structure the web pages and define the layout of different user interface components.
- **CSS** is used for styling the application, making the interface attractive, responsive, and user-friendly.
- **JavaScript** is used for client-side interactivity, validation, and dynamic behavior in the browser.
- **React.js** is used as the main frontend library to build reusable user interface components and manage the single-page application efficiently.

These technologies help in creating a responsive and interactive interface for job seekers and recruiters.

Backend Technologies

The backend of the AI-Based Smart Job Matching and Recruitment System is developed using Node.js and Express.js, ensuring high performance, scalability, and efficient handling of application logic.

• **Node.js** is used to build the server-side environment, enabling fast, event-driven, and scalable application development suitable for real-time features.
• **Express.js** is used as the backend framework to create REST APIs, manage routing, and handle HTTP requests and responses efficiently.

The backend is responsible for core functionalities such as user authentication (JWT-based), job posting and management, application submission and tracking, profile management, real-time communication, and integration with AI services for recommendations and resume analysis. It also ensures secure interaction with the Mongo DB database and smooth data flow across the system.

Database Technologies

The system uses **Mongo DB** as the primary database.

- **Mongo DB** is a No SQL database that stores data in JSON-like document format.
- It is suitable for this project because it provides flexibility, scalability, and efficient handling of large volumes of data.
- It is used to store user profiles, recruiter details, job postings, applications, and AI-related processed data.

Mongo DB integrates effectively with the MERN stack and supports real-time dynamic web applications.

Development Tools

The following tools are used during the development process:

- **Git:**

  Git is used for version control to track code changes, manage project history, and support collaborative development.

- **Git Hub:**

  Git Hub is used as a remote repository platform to store the project code, manage versions, and share the project with team members.

- **Visual Studio Code(VS Code):**

  VS Code is used as the primary code editor for writing, editing, and debugging the project code. It provides extensions and tools that improve productivity in MERN stack development.

- **Postman:**

  Postman is used for API testing to verify the correctness of backend endpoints and ensure smooth communication between frontend and backend.

AI Integration Tools

To enhance the intelligence of the **AI-Based Smart Job Matching and Recruitment System**, AI-related technologies and concepts are integrated into the platform.

• **AI-Based Recommendation Logic:**
The system analyzes user skills, job preferences, and recruiter requirements to suggest relevant job opportunities with improved accuracy.

• **Resume Analysis and Matching:**
AI functionality is used to compare uploaded resumes with job descriptions and calculate matching relevance, helping recruiters identify suitable candidates efficiently.

• **Chat bot / Assistant Support:**
An AI-powered chat bot is integrated to guide users in job searching, profile setup, application processes, and basic career-related queries.

These AI-based enhancements improve the overall user experience and make the recruitment system more efficient, intelligent, and user-friendly.

7. IMPLEMENTATION

The implementation phase of the **AI-Based Smart Job Portal** focuses on converting the system design into a fully functional web application. This phase includes the development of core modules, integration of frontend and backend components, implementation of AI-based features, and testing of the user interface across multiple devices.

The project is developed as a **responsive and modern web application** with animated user interface elements to enhance user experience. It includes separate modules for job seekers, recruiters, and administrators. The system also supports intelligent job recommendations, resume-based matching, secure authentication, and real-time user interactions.

7A. Module Description

The AI-Based Smart Job Portal is divided into several functional modules to ensure proper organization, maintainability, and scalability.

Login Module

The Login Module is one of the most important parts of the system, as it ensures secure access for different types of users such as job seekers, recruiters, and administrators.

Main Features of the Login Module

- User registration with role-based selection
- Secure login using email and password
- JWT-based authentication
- Password encryption for security
- Forgot password / reset password support
- Session handling and protected routes
- Role-based redirection after login

Working Process

When a user enters valid login credentials, the system authenticates the user from the database and generates a secure token. Based on the user role, the system redirects the user to the appropriate dashboard.

Industry-Level Enhancements

- Form validation with proper error messages
- Responsive login page
- Animated background / card effects
- Social login or OTP login can be added in future scope

Admin Panel

The Admin Panel is responsible for monitoring and managing the entire system. It helps control users, recruiters, job posts, applications, and reports.

Main Features of the Admin Panel

- View all registered users and recruiters
- Manage user accounts
- Monitor all job postings
- Remove suspicious or invalid content
- Track total applications and user activity
- Dashboard analytics with charts and counters
- Manage reports and complaints
- System maintenance and moderation

Industry-Level Features

- Modern dashboard cards with statistics
- Graphical charts for user/job/application trends
- Search and filter functionalities
- Responsive admin interface
- Sidebar navigation with icons and animation
- Protected admin-only access

The Admin Panel is designed using a professional dashboard layout to provide better visibility and control over the system.

User Dashboard

The User Dashboard is provided separately for job seekers and recruiters.

Job Seeker Dashboard

The Job Seeker Dashboard allows users to:

- View recommended jobs
- Search and filter job listings
- Update profile details
- Upload resume
- Apply for jobs
- Track application status
- Save jobs for later
- Communicate with recruiters

Recruiter Dashboard

The Recruiter Dashboard allows recruiters to:

- Post new job vacancies
- Edit or delete job posts
- View applicants
- Filter candidates by skills and experience
- Shortlist or reject applications
- Manage company profile
- View analytics of posted jobs

Industry-Level UI Features

- Responsive card-based design
- Animated counters and widgets
- Smooth hover effects
- Search bars with instant filtering
- Status badges for applications
- Profile completion indicators

AI Module

The AI module makes the job portal intelligent and industry-oriented.

Main AI Features

- Smart job recommendation based on skills
- Resume and job description matching
- Candidate ranking system
- AI chat bot support for user guidance
- Personalized job suggestions
- Skill gap analysis for job seekers

Purpose of AI Integration

The purpose of integrating AI is to reduce manual effort, improve job matching accuracy, and enhance the recruitment experience for both candidates and employers

7B. Code Snippets

JWT Authentication Middleware:

```
import jwt from "jsonwebtoken";
export const protect = (req, res, next) => {
  const token = req.headers.authorization?.split(" ")[1];
  if (!token) {
    return res.status(401).json({ message: "Unauthorized access" });
  }
  try {
    const decoded =jwt.verify(token,
    process.env.JWT_SECRET); req.user = decoded;
    next();
  } catch (error) {
    res.status(401).json({ message: "Invalid token" });
  }
};
```

Explanation

This middleware protects private routes by checking whether the user has a valid token. It is used in dashboards, job posting routes, and profile-related APIs.

Login Controller:

```
import User from
"../models/User.js"; import
bcrypt from "bcryptjs"; import
jwt from "jsonwebtoken";

export const loginUser = async (req, res)
  => { try {
    const { email, password } = req.body;

    const user = await User.findOne({
    email }); if (!user) {
      return res.status(400).json({ message: "User not found" });
    }

    const isMatch = await bcrypt.compare(password,
    user.password); if (!isMatch) {
      return res.status(400).json({ message: "Invalid credentials" });
    }

    const token = jwt.sign(
      { id: user._id, role: user.role },
      process.env.JWT_SECRET,
      { expiresIn: "7d" }
    );

    res.status(200).json({
      message: "Login
      successful", token,
      user,
    });
  } catch (error) {
    res.status(500).json({ message: "Server error" });
  }
};
```

Explanation

This code validates login credentials, compares passwords securely, and generates a JWT token for authenticated access.

Job Posting API:

```
import Job from "../models/Job.js";

export const createJob = async (req, res) => {
  try {
    const job = await Job.create({
      title: req.body.title,
      company: req.body.company,
      location: req.body.location,
      skills: req.body.skills,
      description: req.body.description,
      recruiter: req.user.id,
    });

    res.status(201).json(job);
  } catch (error) {
    res.status(500).json({ message: "Job creation failed" });
  }
};
```

Explanation

This API allows recruiters to create new job postings. The posted job is linked with the recruiter ID.

AI Job Recommendation Logic:

```
export const recommendJobs = (userSkills, jobs) => {
  return jobs.map((job) => {
    const matchedSkills = job.skills.filter((skill) =>
      userSkills.includes(skill)
    );

    const score = (matchedSkills.length / job.skills.length) * 100;

    return {
      ...job,
      matchScore: score.toFixed(2),
    };
  }).sort((a, b) => b.matchScore - a.matchScore);
};
```

<u>Explanation</u>

This logic compares user skills with job skills and generates a matching score. Jobs with higher scores are recommended to the user first.

React Dashboard Card Example:

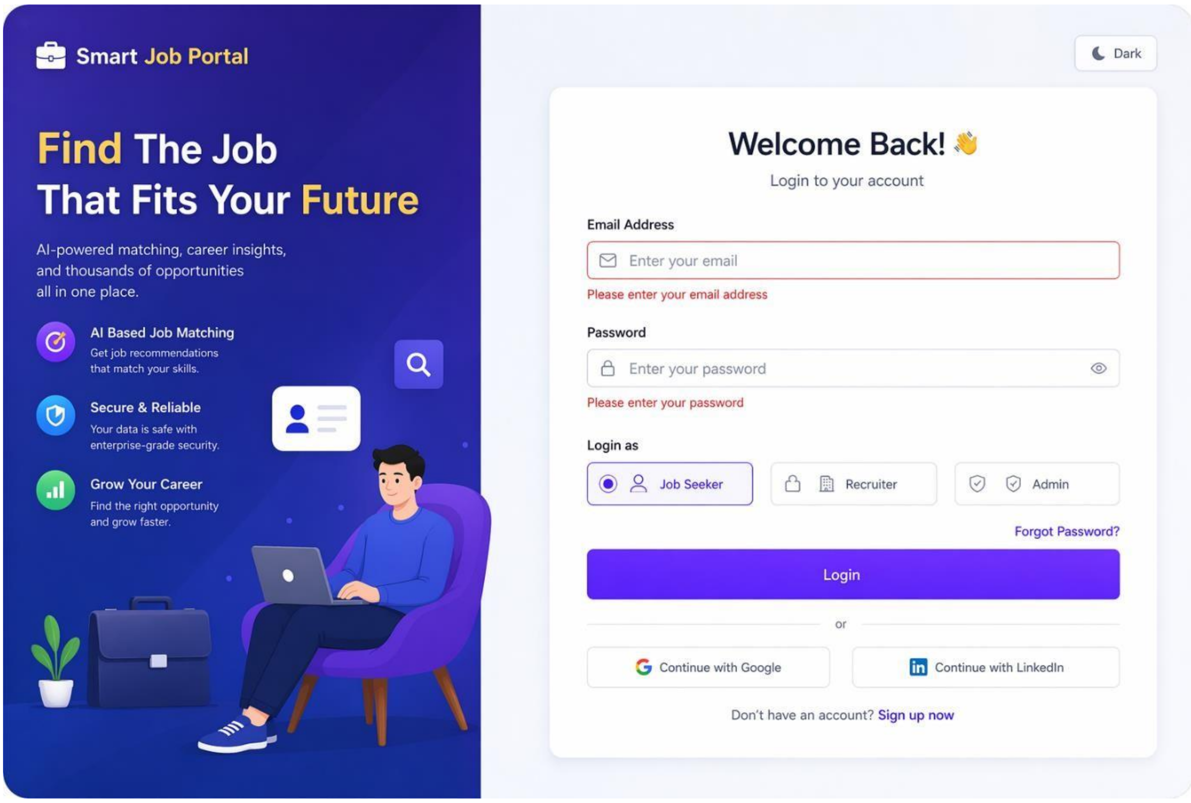
```
const DashboardCard = ({ title, count, icon }) => {

  return (

    <div className="bg-white shadow-lg rounded-2xl p-5 hover:scale-105 transition duration-
300">

      <div className="flex items-center justify-between">

        <div>

          <h3 className="text-gray-500 text-sm">{title}</h3>

          <h2 className="text-2xl font-bold">{count}</h2>

        </div>

        <div className="text-blue-600 text-3xl">{icon}</div>

      </div>

    </div>

  );

};



export default DashboardCard;
```

<u>**Explanation**</u>

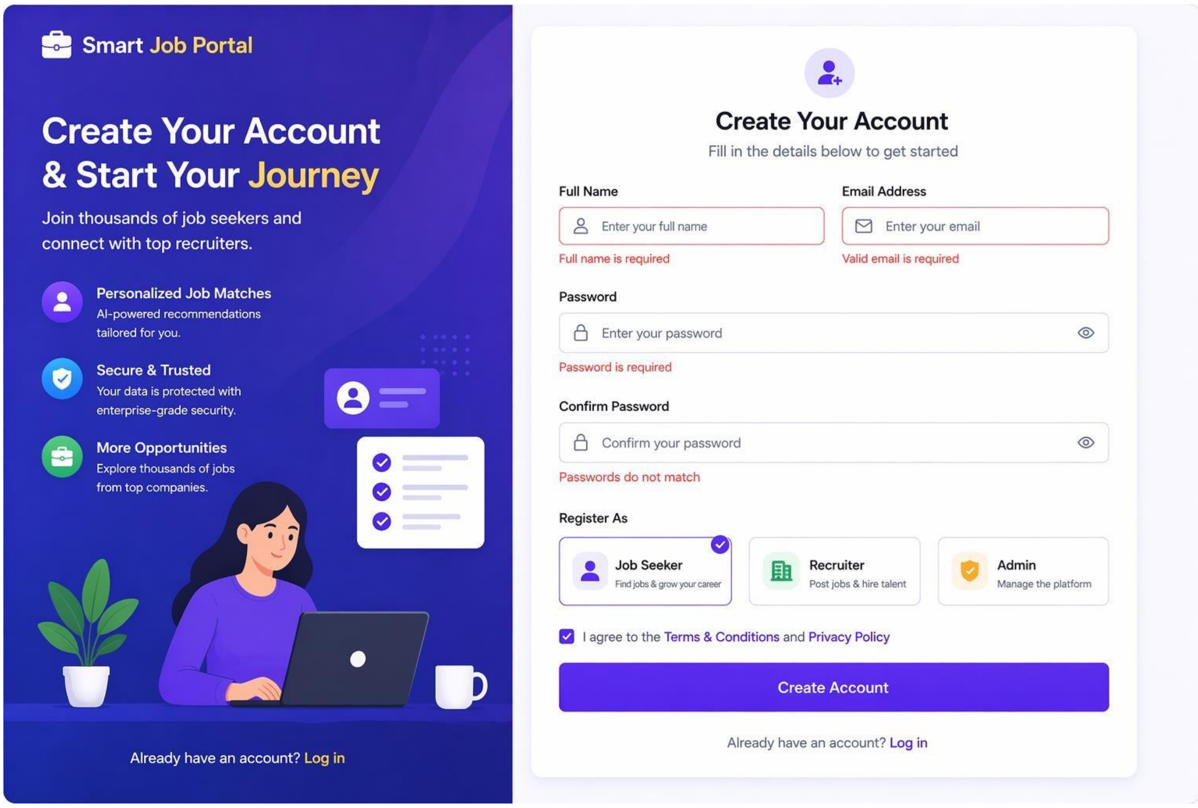
This component is used in modern dashboard design to show quick statistics such as total jobs, total applications, and total users

7C. Screenshots

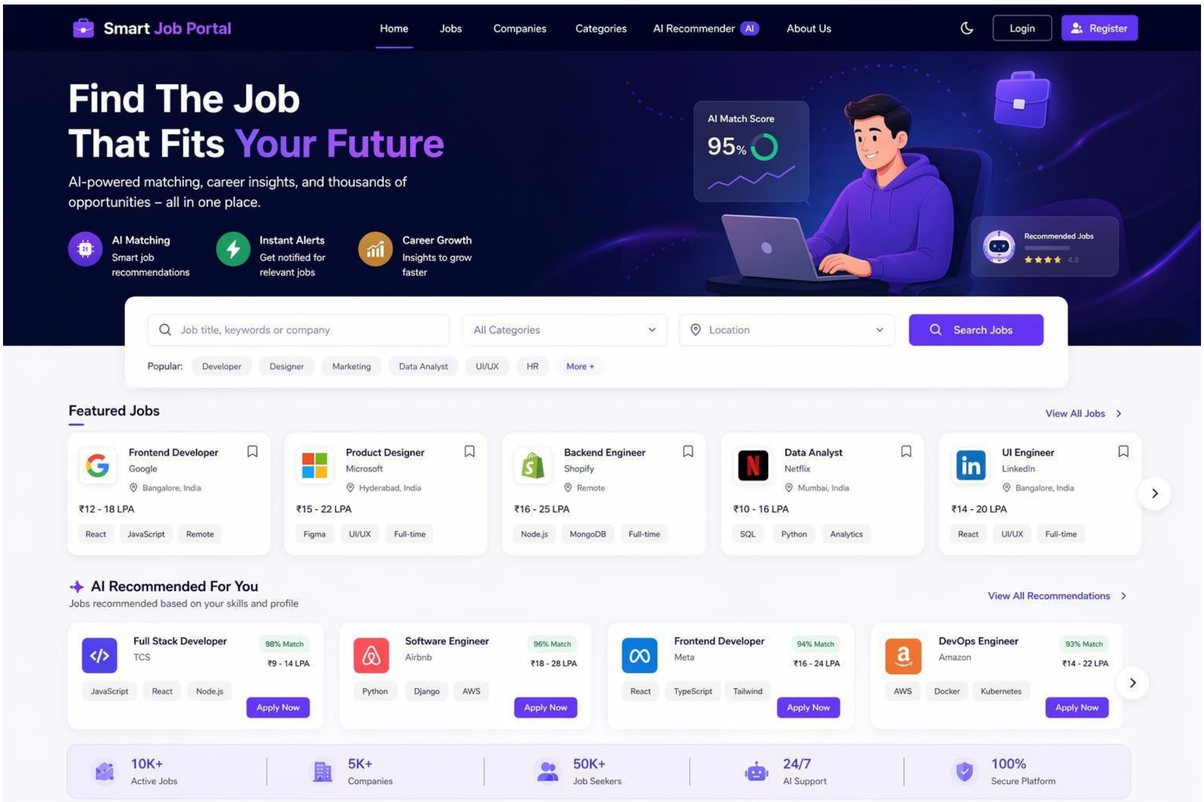
LOGIN PAGE:



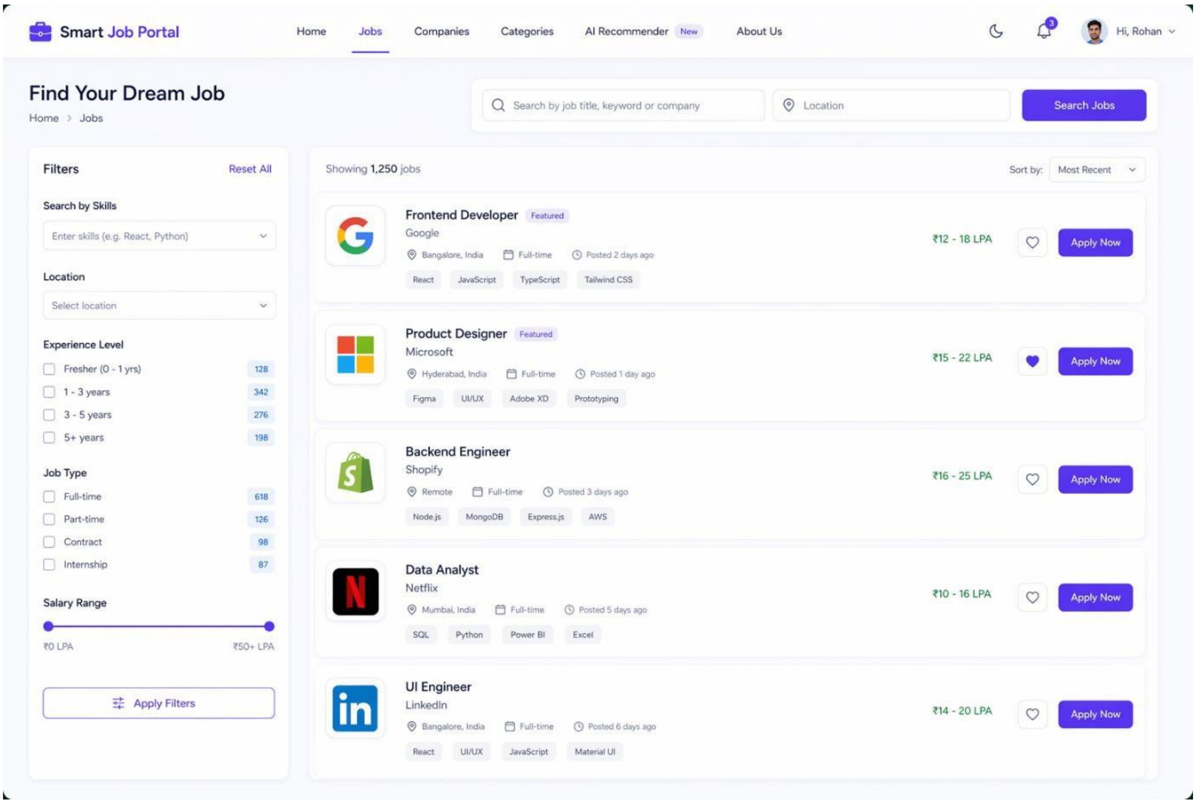
REGISTRATION PAGE:



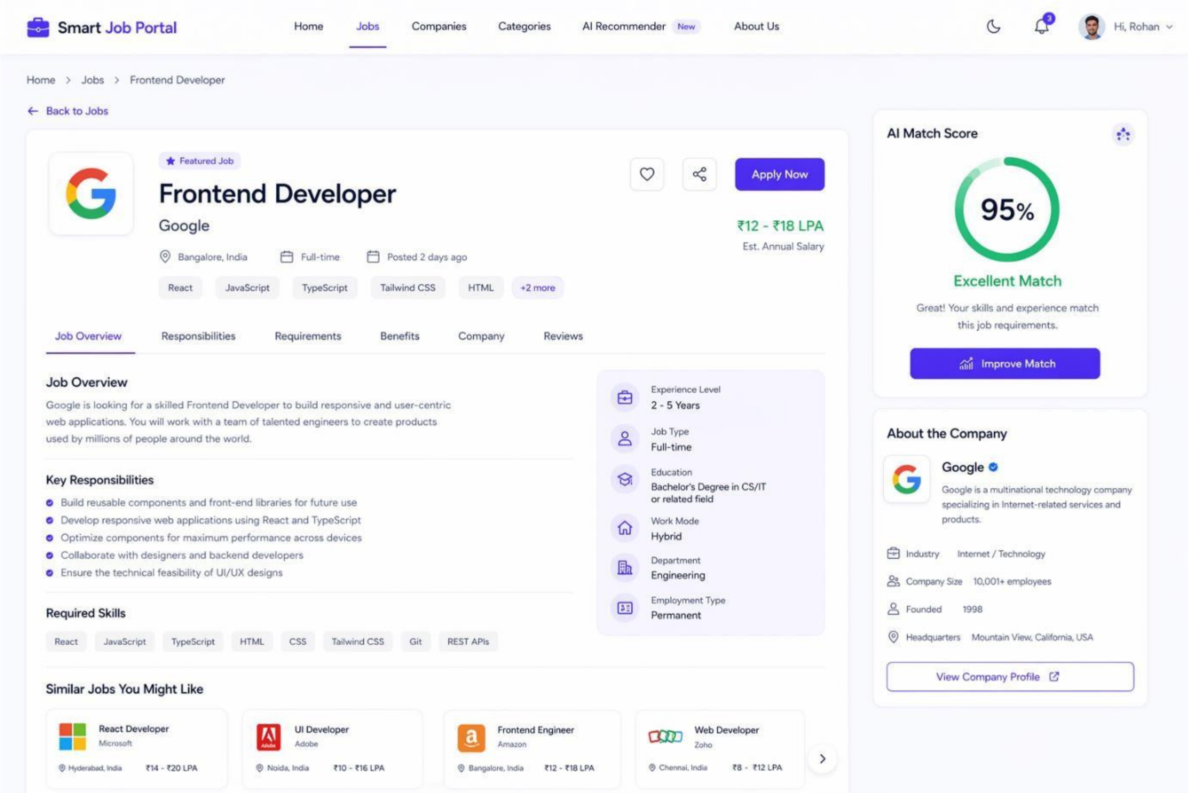
HOME PAGE:


Smart Job Portal
Home
Jobs
Companies
Categories
AI Recommender AI
About Us
Login
Register
Find The Job
That Fits Your Future
AI-powered matching, career insights, and thousands of opportunities – all in one place.
AI Matching
Smart job recommendations
Instant Alerts
Get notified for relevant jobs
Career Growth
Insights to grow faster
AI Match Score
95%
Recommended Jobs
Job title, keywords or company
All Categories
Location
Search Jobs
Popular:
Developer
Designer
Marketing
Data Analyst
UI/UX
HR
More +
Featured Jobs
View All Jobs
Frontend Developer
Google
Bangalore, India
₹12 - 18 LPA
React
JavaScript
Remote
Product Designer
Microsoft
Hyderabad, India
₹15 - 22 LPA
Figma
UI/UX
Full-time
Backend Engineer
Shopify
Remote
₹16 - 25 LPA
Node.js
MongoDB
Full-time
Data Analyst
Netflix
Mumbai, India
₹10 - 16 LPA
SQL
Python
Analytics
UI Engineer
LinkedIn
Bangalore, India
₹14 - 20 LPA
React
UI/UX
Full-time
AI Recommended For You
Jobs recommended based on your skills and profile
View All Recommendations
Full Stack Developer
TCS
98% Match
₹9 - 14 LPA
JavaScript
React
Node.js
Apply Now
Software Engineer
Airbnb
96% Match
₹18 - 28 LPA
Python
Django
AWS
Apply Now
Frontend Developer
Meta
94% Match
₹16 - 24 LPA
React
TypeScript
Tailwind
Apply Now
DevOps Engineer
Amazon
93% Match
₹14 - 22 LPA
AWS
Docker
Kubernetes
Apply Now
10K+
Active Jobs
5K+
Companies
50K+
Job Seekers
24/7
AI Support
100%
Secure Platform

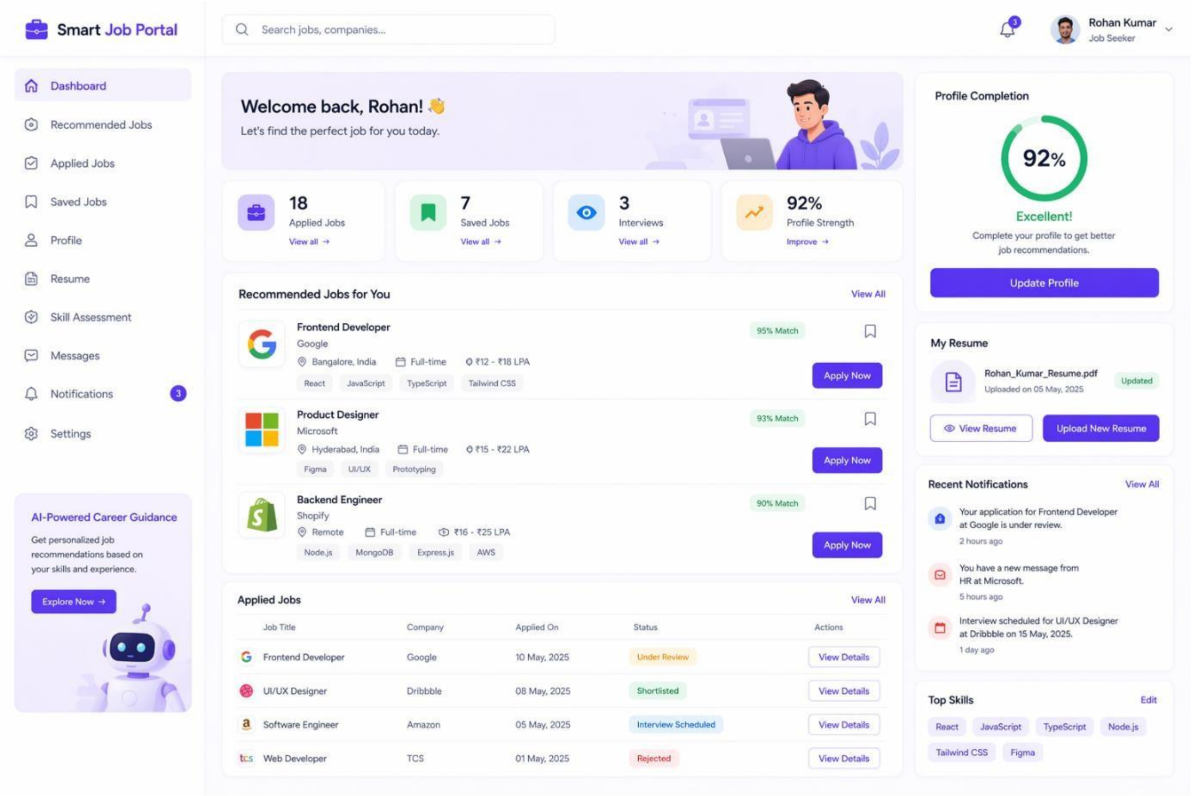
JOB LISTING PAGE:



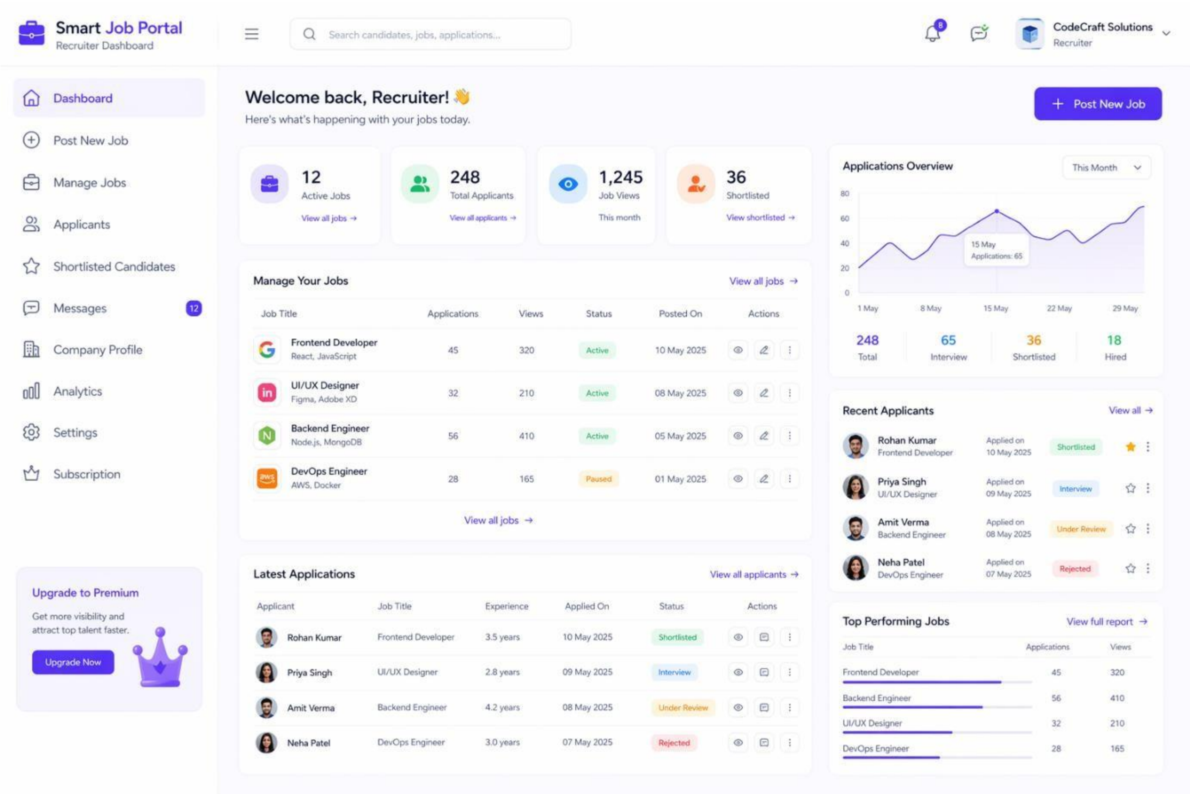
JOB DETAILS PAGE:



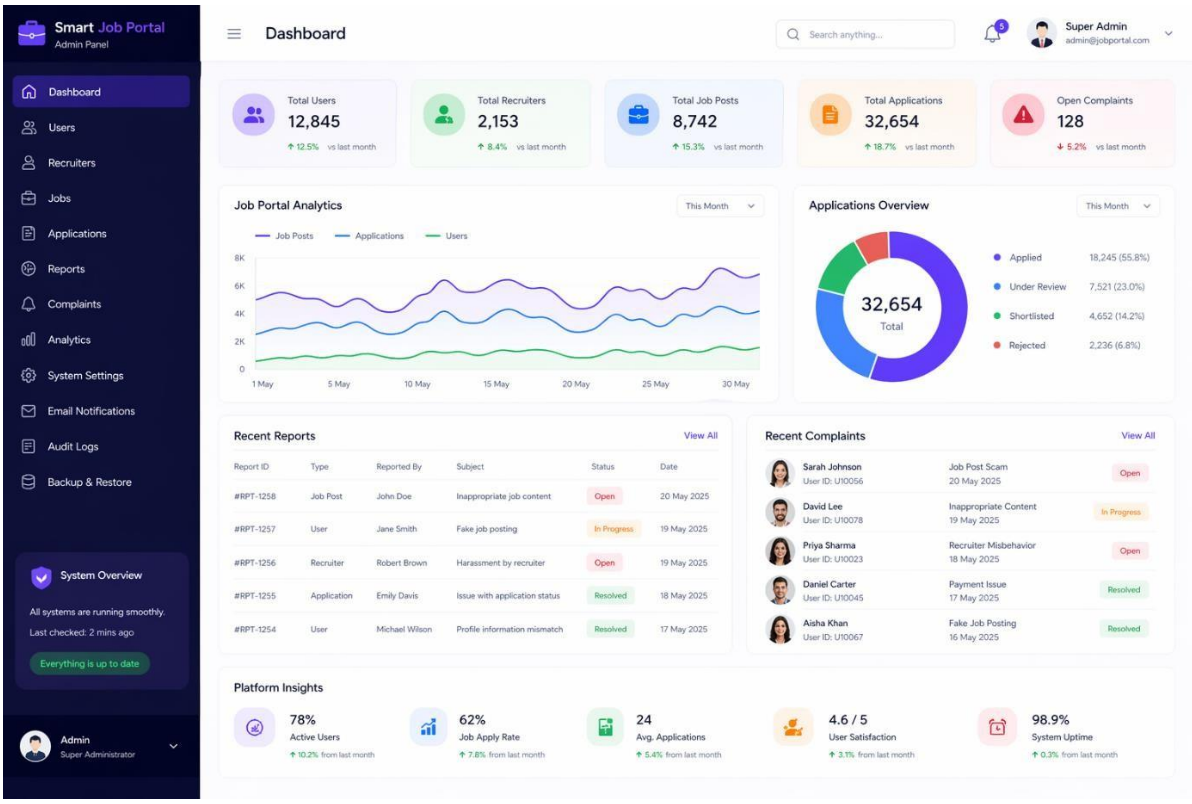
JOB SEEKER DASHBOARD:



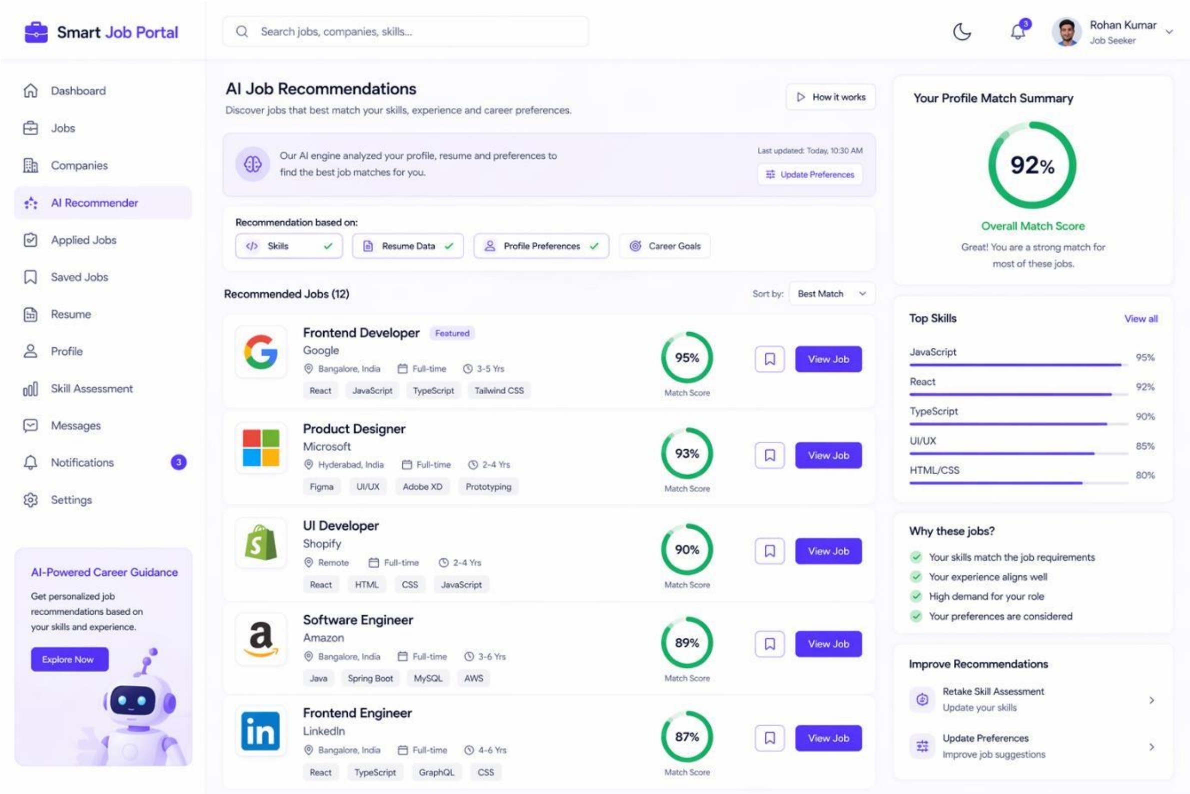
RECRUITER DASHBOARD:



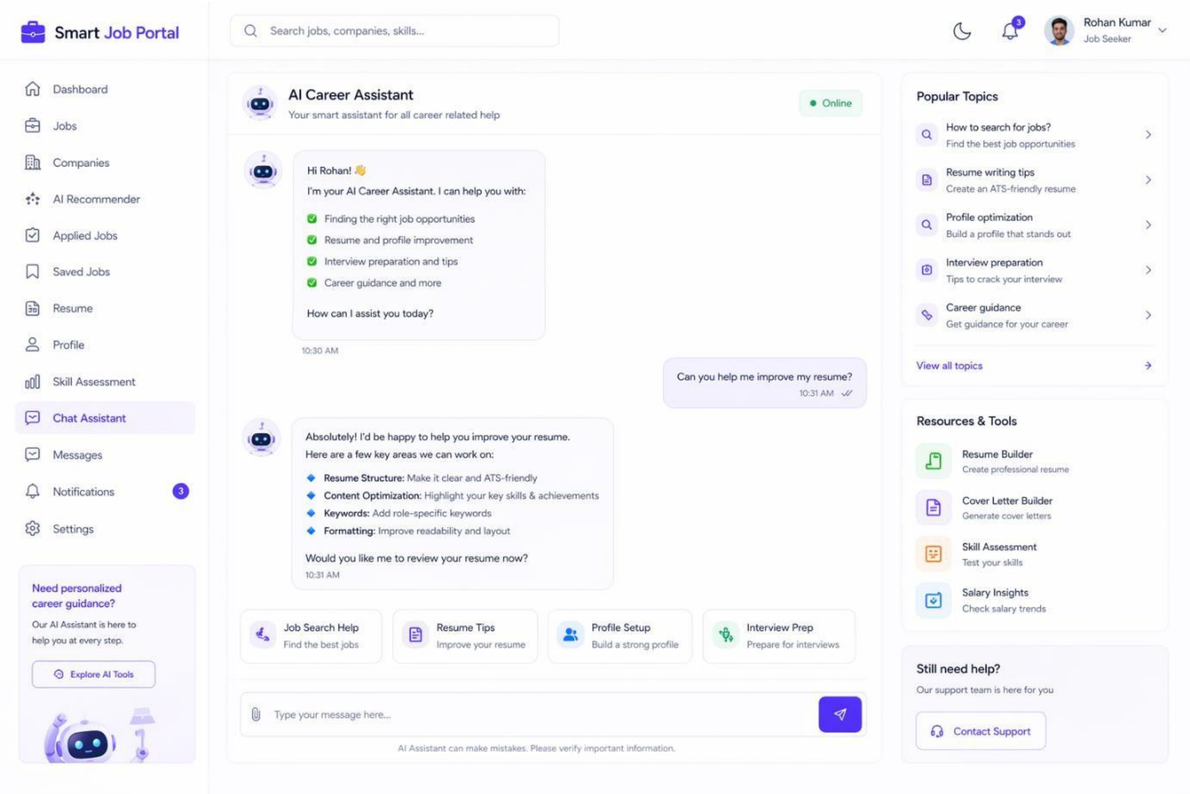
ADMIN DASHBOARD:



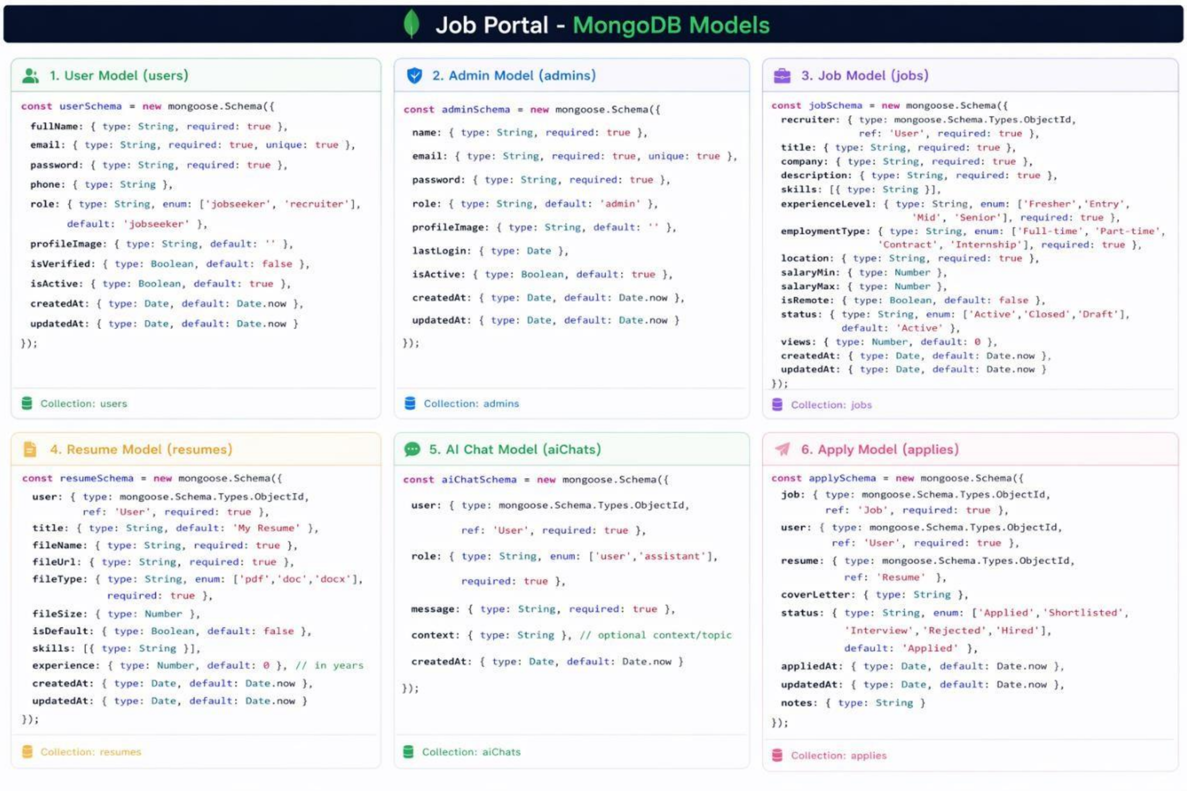
AI Recommendation PAGE:



CHAT BOAT PAGE:



MONGODB JOB PORTAL SCHEMA SCREENSHOT:



8. TESTING

Testing is one of the most important phases in software development. It ensures that the system functions correctly, meets user requirements, and performs reliably under different conditions. In the **AI-Based Smart Job Matching and Recruitment System**, testing is carried out to verify the proper working of all modules such as user authentication, job posting, AI-based job recommendation, resume upload, chat bot assistance, and dashboard management.

The testing process helps identify bugs, validate system behavior, and improve the overall quality of the application. Both **Unit Testing** and **Integration Testing** are used to confirm that the individual components and the complete system work smoothly

Test Cases Table

The following table shows some important test cases performed on the Smart Job Portal system

Test Case ID	Test Case	Input	Expected Output	Result
TC01	User Login with valid credentials	Valid email and password	User should log in successfully and be redirected to dashboard	Pass
TC02	User Login with invalid credentials	Invalid email or password	Error message should be displayed	Pass
TC03	User Registration	Full name, email, password, role	New account should be created successfully	Pass
TC04	Password mismatch during registration	Different password and confirm password	Validation message should be shown	Pass
TC05	Job Search	Keyword: “Frontend Developer”	Relevant job listings should be displayed	Pass
TC06	Apply for Job	Selected job and user profile	Job application should be submitted successfully	Pass
TC07	Resume Upload	PDF/DOC file	Resume should be uploaded and stored successfully	Pass
TC08	AI Recommendation	User skills and profile data	Matched jobs should be recommended with score	Pass
TC09	Recruiter posts a job	Job title, company, location, skills	Job should be posted successfully	Pass
TC10	Recruiter edits job	Updated job details	Job details should be updated in database	Pass
TC11	Recruiter views applicants	Job ID	Applicant list should be displayed	Pass
TC12	Admin views dashboard statistics	Dashboard request	Total users, recruiters, jobs, and applications should be shown	Pass
TC13	Chat bot assistance	User query for resume/interview/job help	Assistant should return useful guidance	Pass
TC14	Protected route access without token	No token	Unauthorized access message should be shown	Pass
TC15	Profile update	Updated name, skills, preferences	Profile should be updated successfully	Pass

8A. Unit Testing

Unit Testing is performed to test individual components or modules of the system separately. In this method, each function or module is checked independently to confirm that it performs its intended task correctly.

Examples of Unit Testing in this Project

- Testing login function with valid and invalid credentials
- Testing registration form validation
- Testing job posting API
- Testing AI recommendation function
- Testing resume upload function
- Testing chat bot response function

Advantages of Unit Testing

- Helps identify bugs at an early stage
- Improves code reliability
- Makes debugging easier
- Ensures each module works independently

In this project, Unit Testing is mainly used for backend controllers, API routes, validation logic, and frontend component behavior.

8B. Integration Testing

Integration Testing is used to verify whether different modules of the system work together properly after integration. It ensures smooth communication between frontend, backend, database, and AI modules.

Examples of Integration Testing in this Project

- Checking whether the login form successfully communicates with the backend API
- Verifying that job search results are fetched from the database and displayed on the frontend
- Testing whether job application data is stored correctly after form submission
- Confirming that AI recommendation results are generated using user profile and job database information
- Ensuring the recruiter dashboard correctly displays applicants fetched from the backend
- Verifying chat bot interaction with user input and response module

Advantages of Integration Testing

- Detects communication problems between modules
- Validates complete workflow of the application
- Ensures API and database integration works correctly
- Improves overall system reliability

In this project, Integration Testing is essential because the Smart Job Portal consists of multiple connected modules such as authentication, job management, AI recommendation, and chat bot support.

8C. Functional Testing

Functional Testing is performed to check whether the system functionalities work according to the requirements.

Functional areas tested

- User registration and login
- Role-based access
- Job posting and management
- Job application process
- Resume upload
- Job recommendation
- Dashboard operations
- Admin monitoring functions

This testing confirms that the system behaves correctly from the user's perspective.

8D. User Interface Testing

User Interface Testing is used to verify the visual design and usability of the application.

Key areas tested

- Responsive layout on different screen sizes
- Button alignment and navigation
- Form field visibility and validation messages
- Sidebar menu functionality
- Dashboard cards, charts, and tables
- Animated UI components

This testing ensures that the application is user-friendly, modern, and professional in appearance.

8E. Performance Testing

Performance Testing checks whether the system performs efficiently under normal and heavy loads.

<u>Performance aspects tested</u>

- Fast loading of dashboard pages
- Quick search and filter response
- Job listing rendering performance
- API response time
- Database query performance
- Simultaneous access handling

This helps ensure that the portal can support multiple users without major slowdown

8F. Testing Summary

The testing results show that the **AI-Based Smart Job Matching and Recruitment System** performs effectively across its major modules. Login, registration, job search, application management, recruiter actions, admin monitoring, and AI-based recommendations all function correctly according to the expected results.

The use of **Unit Testing** and **Integration Testing** helps improve system quality, reduce errors, and ensure that the project is reliable for real-world usage. Therefore, the testing phase confirms that the application is ready for deployment and further enhancement.

9. RESULTS & DISCUSSION

The **AI-Based Smart Job Matching and Recruitment System** has been successfully designed and implemented using the MERN Stack (Mongo DB, Express.js, React.js, Node.js), integrated with AI-based recommendation and chatbot features.

The system provides a modern, responsive, and user-friendly interface with efficient backend processing and intelligent job matching capabilities.

9A. Output Screenshots

This section presents the output of different modules of the system through screenshots. These outputs demonstrate the working functionality, UI/UX design, and real-time performance of the application.

Login & Registration Output

- Secure login system with **JWT authentication**
- Role-based access (**Job Seeker / Recruiter / Admin**)
- Responsive UI with modern card design
- Validation messages for incorrect inputs

Output:

- Users can log in successfully and access their dashboards
- New users can register with proper validation

Home Page Output

- Hero section with animated UI
- Job search functionality
- Featured job listings
- AI-based recommendations

Output:

- Users can search jobs quickly
- AI suggests relevant jobs based on profile

Job Listing & Job Details Output

- Filter by skills, location, experience
- Job cards with clean UI
- Detailed job information page

Output:

- Users can view jobs easily
- Detailed job page shows full information with **Apply button and match score**

Job Seeker Dashboard Output

- Applied jobs list
- Recommended jobs
- Resume upload
- Profile management
- Notifications

Output:

- Users can track application status
- AI recommendations improve job matching

Recruiter Dashboard Output

- Post new job
- Manage job listings
- View applicants
- Shortlist candidates

Output:

- Recruiters can efficiently manage hiring process
- Real-time applicant tracking system

Admin Dashboard Output

- Total users
- Total recruiters
- Total job posts
- Total applications
- Analytics charts

Output:

- Admin can monitor full system
- Reports and complaints can be handled easily

AI Recommendation System Output

- AI-based job matching
- Skill-based filtering
- Resume-based analysis

Output:

- Jobs are recommended with **matching percentage (%)**
- Improves user experience and job accuracy

Chat bot / Assistance Output

- AI chat bot interface
- Job guidance
- Resume suggestions
- Interview preparation tips

Output:

- Users receive instant assistance
- Improves engagement and usability

9B. Performance Analysis

The system performance has been evaluated based on speed, responsiveness, and efficiency.

Response Time

- API response time: **Fast (under 1–2 seconds)**
- Database query execution: **Optimized using Mongo DB indexing**
- UI rendering: **Smooth with**

React.js Scalability

- Mongo DB allows handling large data efficiently
- Node.js supports multiple concurrent users
- Cloud deployment (AWS/Render) ensures scalability

Efficiency

- AI model improves job matching accuracy
- Reduced manual effort in job searching
- Efficient recruiter workflow

Responsiveness

- Fully responsive design
- Works on:
    - Desktop
    - Tablet
    - Mobile devices

Security Performance

- JWT authentication for secure login
- Password hashing (bcrypt)
- Role-based access control

9C. Observations

During the development and testing of the Smart Job Portal, several important observations were made

System Functionality

- All modules work correctly as expected
- Smooth interaction between frontend and backend
- API integration is stable and efficient

User Experience

- Clean and modern UI improves usability
- Easy navigation for all user roles
- AI recommendations enhance user engagement

AI Performance

- AI model successfully suggests relevant jobs
- Resume-based matching improves accuracy
- Chat bot provides useful real-time assistance

Limitations

- AI recommendations depend on quality of user data
- Real-time updates may need Web Socket optimization
- Chat bot responses can be improved with advanced NLP

Future Improvements

- Add **real-time notifications (Socket.IO)**
- Improve AI model using **Machine Learning (ML models)**
- Add **video interview system**
- Integrate **advanced analytics dashboard**

9D. Conclusion of Results

The developed **AI-Based Smart Job Portal** meets all the project requirements and demonstrates strong performance in terms of functionality, usability, and efficiency.

The integration of **AI recommendation and chat bot system** significantly enhances the platform, making it more intelligent and user-friendly compared to traditional job portals.

10. CONCLUSION

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The **AI-Based Smart Job Matching and Recruitment System** has been successfully developed and implemented using the MERN Stack (Mongo DB, Express.js, React.js, Node.js) along with intelligent AI-based features. This system provides a modern, scalable, and efficient solution for connecting job seekers with recruiters in a streamlined and user-friendly manner.

10A. What We Achieved

Through this project, the following objectives have been successfully accomplished:

- Developed a **full-stack web application** using modern technologies
- Implemented **secure authentication system** using JWT
- Designed **role-based access control** for Job Seekers, Recruiters, and Admin
- Created a **responsive and modern UI** with smooth user experience
- Built **dynamic job search and filtering system**
- Enabled **job application and tracking functionality**
- Developed **recruiter panel** for job posting and applicant management
- Implemented **admin dashboard** for system monitoring and analytics
- Integrated **AI-based job recommendation system** for better matching
- Developed an **AI chat bot assistant** for user guidance and support
- Ensured **database efficiency and scalability** using Mongo DB

10B. Summary of the System

The Smart Job Portal is a complete online platform that simplifies the recruitment process by combining automation, intelligent recommendations, and modern UI design.

For Job Seekers

- Register and log in securely
- Search and apply for jobs
- Upload resumes and manage profiles
- Receive AI-based job recommendations
- Track application status in real-time

For Recruiters

- Post and manage job listings
- View and shortlist candidates
- Analyze applicant data
- Manage hiring process efficiently

For Admin

- Monitor overall system performance
- Manage users and recruiters
- Track job postings and applications
- Handle reports and system activities

10C. Overall Outcome

The system successfully meets all the functional and technical requirements of a modern job portal. The integration of AI features enhances the efficiency of job matching and improves user engagement. The platform is scalable, secure, and ready for real-world deployment.

The project demonstrates strong practical implementation of:

- Full Stack Development (MERN)
- API Integration
- Database Design
- AI-based Features
- Modern UI/UX Design

10D. Final Statement

In conclusion, the **AI-Based Smart Job Matching and Recruitment System** is an advanced, industry-level solution that improves the traditional recruitment system by incorporating automation, intelligence, and user-centric design. It provides a strong foundation for further enhancements and real-world applications.

11. FUTURE SCOPE

The **AI-Based Smart Job Matching and Recruitment System** developed in this project provides a strong foundation for modern recruitment solutions. However, there are several opportunities to further enhance the system by integrating advanced AI technologies, improving performance, and expanding platform accessibility for better user experience and scalability.

11A. Improvements

The current system can be improved in the following ways to make it more powerful and industry-ready:

- **Real-time Notifications**
  - Implement **Socket.IO / Web Sockets** for instant updates
  - Notify users about job applications, interview calls, and messages
- **Advanced Search & Filters**
  - Add filters like salary range, company rating, job type
  - Improve search using **Elastic Search** for faster and accurate results
- **Enhanced Security**
  - Implement **Two-Factor Authentication (2FA)**
  - Add **OAuth login (Google, LinkedIn)**
  - Improve API security with rate limiting
- **Performance Optimization**
  - Use **caching (Redis)** for faster response
  - Optimize database queries and indexing
  - Implement lazy loading and code splitting in React
- **Real-Time Chat System**
  - Enable direct communication between recruiter and job seeker
  - Add file sharing (resume, documents)
- **Interview System**
  - Integrate **video interview feature (WebRTC / Zoom API)**
  - Schedule interviews within the portal

11B. AI Integration (Advanced Level)

The system already includes basic AI features, but it can be enhanced further using advanced AI and Machine Learning techniques:

- **Smart Resume Screening**
  - Automatically analyze resumes using NLP
  - Rank candidates based on job requirements
- **AI Job Recommendation Engine**
  - Use **Machine Learning models** to improve accuracy
  - Personalized recommendations based on user behavior
- **AI Chat bot (Advanced)**
  - Upgrade chat bot using **LLM (Open AI / Groq AI)**
  - Provide intelligent responses for job queries, interview preparation
- **Skill Gap Analysis**
  - Suggest skills users need to improve for better job matching
  - Recommend courses or certifications
- **Interview Preparation AI**
  - Generate real-time interview questions
  - Provide mock interview simulation
- **Predictive Analytics**
  - Predict hiring trends
  - Analyze job market demand

11C. Mobile Application Version

To improve accessibility and user engagement, the system can be extended into a mobile application:

- **Cross-Platform Mobile App**
  - Develop using **React Native / Flutter**
  - Available for both Android and IoS
- **Mobile Features**
  - Push notifications for job alerts
  - Easy job application with one click
  - Resume upload from mobile
  - Chat bot assistance on mobile
- **Offline Support**
  - Save jobs and view later without internet
  - Sync data when online
- **Improved User Experience**
  - Touch-friendly UI
  - Faster navigation
  - Personalized dashboard

11D. Cloud & Deployment Enhancements

- Deploy on **AWS / Azure / GCP**
- Use **Docker & Kubernetes** for scalability
- Implement **CI/CD pipelines (GitHub Actions)**
- Use **CDN for faster content delivery**

11E. Future Expansion

- Multi-language support
- Global job marketplace integration
- Company rating and review system
- Block chain-based certificate verification
- Freelance job marketplace integration

12. REFERENCES

The following references were used during the development and documentation of the **AI-Based Smart Job Portal**. These include books, research papers, and online resources that helped in understanding the technologies, methodologies, and implementation strategies.

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